

Malhar Mangtani

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Portfolio: malharmangtani.github.io

Educational Information

Columbia University, New York City, USA M.S. in Computer Science

Expected Dec 2027

- Admitted for Fall 2026

Sarvajanik College of Engineering and Technology, Surat, India B. Tech in Computer Engineering

Sep 2022– May 2026

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Machine Vision, Python for Data Science, Data Structures & Algorithms, Database Management Systems, Operating Systems, Cloud Computing, Compiler Design.
- Cumulative Performance Index (CPI): **9.35/10.00** | Degree CGPA: **9.11/10.00**

Sarvajanik College of Engineering and Technology, Surat, India Honors in Artificial Intelligence/ Machine Learning

Dec 2023 – Dec 2025

- Relevant Coursework: Introduction to Artificial Intelligence and Robotics, Machine Learning and Applications, Deep Learning, Modern trends in AI-ML.
- CGPA: **10.00/10.00**

Laurentian University, Sudbury, Canada International Experience Program (IEP)

Jul–Aug 2024

- Relevant Coursework: Mobile Computing and Wireless Communication, Machine Learning.

Work Experience

Oklahoma State University, Industrial Training and Assessment Center (ITAC), USA

Feb–Apr 2026

AI Intern (Hybrid)

- Developed SmartITAC — an AI chatbot enabling the ITAC team to query 167,000+ energy assessment records through plain English, reducing a 1–2 hour task to seconds.
- Built an end-to-end RAG pipeline using HuggingFace embeddings, ChromaDB, and Groq LLM API with a multi-mode query system supporting keyword search, geographic filtering, statistical aggregation, and time-based filtering.
- Built a FastAPI backend with real-time streaming and an HTML chat frontend accessible to all team roles.
- Received official internship completion letter from Dr. Hitesh Vora, Oklahoma State University, funded by U.S. Department of Energy (DOE).

USICAMM, Secretaría de Educación Pública (SEP), Mexico

Jan–Jul 2025

AI Intern (Remote)

- Contributed to an AI-driven teacher management platform for registration, promotion, and recognition.
- Built AI-based background check, profile evaluation, and validation search engine modules.
- Received official experience letter from USICAMM (SEP, Mexico) for significant contributions.

Laurentian University, Sudbury, Canada

Jul–Aug 2024

AI/ ML Research Intern (On-site)

- Built a DeepLabV3 + ResNet50 model for nuclei segmentation and predictive analysis in histopathology images.
- Achieved high precision, recall, and F1-scores, supporting reliable diagnostic workflows.
- Tools:** Python, TensorFlow, DeepLabV3, ResNet50

Research Projects and Presentations

Synthetic CT Generation from MRI

2025

- Built and iteratively improved a deep learning pipeline for MRI-to-CT synthesis, progressing from a 3D Pix2Pix GAN (3D U-Net generator) to an anatomy-aware SwinGAN (SwinUNETR generator), both paired with a ConvNeXt-based anatomical classifier.
- Achieved stronger results with SwinGAN across all anatomical regions (avg. MS-SSIM **0.932**, PSNR **27.83** dB, MAE **0.027**), with peak performance on Thorax (MS-SSIM **0.947**, PSNR **28.30** dB, MAE **0.026**), outperforming the Pix2Pix baseline with high structural fidelity in complex bone-air interface regions.

Malaria Parasite Detection

2025

- Developed a malaria parasite detection system in thin blood smear images using ConvNeXt with Grad-CAM++ for explainable malaria detection.
- Resulted in an IEEE conference publication and was presented at ICETCI 2025, Hyderabad, India.
- Received the Best Presenter Award in my session.

Emotion Detection Using Deep Learning Algorithms

2025

- Developed and evaluated multiple CNN architectures (ResNet50, EfficientNetB3, VGG19) for a 6-class image classification task.
- Achieved highest performance with ResNet50, reaching **93.36%** validation accuracy and **67.22%** test accuracy, supported by confusion matrices and detailed classification reports.

American Sign Language Recognition System

2024

- Developed a real-time sign language recognition system using LSTMs with MediaPipe and OpenCV for hand and gesture tracking.
- Enabled accurate gesture-to-text translation to improve communication accessibility.
- Achieved **99%** training accuracy, with testing accuracy of **96.49%** in letter recognition, **99.995%** in number recognition, and **98.396%** in static word recognition, obtaining an average of **98.29%** based on gesture recognition.

Real-Time Face and Hand Recognition with Finger Count

2024

- Implemented a recognition system integrating facial feature detection and hand gesture recognition.
- Developed a finger-counting module for real-time human-computer interaction.

Publications

Malaria Parasite Detection in Thin Blood Smear Images Using ConvNeXt with Explanation

2025

- Published in the Proceedings of the IEEE International Conference on Emerging Techniques in Computational Intelligence (ICETCI 2025), Hyderabad, India. DOI: 10.1109/ICETCI67340.2025.11258060

Technical Skills and Interests

- **Programming Languages:** Python, Java
- **Natural Language Processing & Generative AI:** Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Prompt Engineering, HuggingFace Transformers, Sentence Embeddings, Vector Databases (ChromaDB), Groq API, LangChain
- **Computer Vision and Medical Imaging:** OpenCV, MONAI, SimpleITK/ITK, Image Processing, Image Augmentation, MRI/CT Analysis
- **Tools and Frameworks:** FastAPI, MediaPipe, Jupyter Notebook, Conda / Virtual Environments, SQL, VS Code
- **Specialized Models:** GANs, CNNs, RNNs, LSTMs, Transfer Learning, Transformer Models
- **Machine Learning and Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Model Evaluation Metrics (Precision, Recall, F1-score, ROC-AUC), Kaggle
- **Areas of Interest:** Artificial Intelligence and Machine Learning, Deep Learning & Computer Vision, Natural Language Processing & Conversational AI, Retrieval-Augmented Generation, Data Science & Predictive Analytics, Explainable and Trustworthy AI, Generative AI (Diffusion Models, GANs, VAEs), Medical Imaging and Healthcare AI